ATHANASIOS GOURDOMICHALIS
Java Programms
App2.java

A. The method:

```
int factorial(int n)
```

calculates and returns the factorial of a positive integer or zero ($n \geq 0$), as follows:

if $\mathbf{n = 0}$:
    $\mathbf{n! = 1}$
else:
    $\mathbf{n! = 1 \cdot 2 \cdot \ldots \cdot n}$

B. The method:

```
int combinations(int n, int k)
```

calculates and returns the number of combinations of $\boldsymbol{n}$ elements taken $\boldsymbol{k}$ at a time, as follows:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

where $\boldsymbol{n}$ and $\boldsymbol{k}$ are positive integers and $\mathbf{n \geq k}$.

The combinations method calls the factorial method to calculate the factorial where required.

C. Write a program (**`class App2`**) that includes the two methods (**`factorial`** and **`combinations`**) and additionally a main method (main method) that does the following:

1. Prompts the user for and reads the integer $\boldsymbol{n}$.
2. Prompts the user for and reads the integer $\boldsymbol{k}$.
3. Checks the validity of the data based on the following rules:

- **3.1.** $\boldsymbol{n}$ and $\boldsymbol{k}$ must be positive integers $(\boldsymbol{n > 0, k > 0})$, and
- **3.2.** $\boldsymbol{n \geq k}$.

In case the user does not provide valid data, the program must **terminate** its execution and display an appropriate message.

4. If the data is valid, the program **calculates and displays** the number of combinations of n choose k. The calculation of combinations must be done by calling the combinations method.

When calculating the factorial, ignore the case of integer overflow for the variable storing the result.

Complete the code in the file **App2.java.**

Every program must compile and run from the command line (not through an IDE).